Developing RSE skills at the novice level

Student RSE model for research support, experiential learning, and technical workforce development

Paula Lackie, Oliver Hall, Tim Dennis, Leigh Phan
Carleton College DataSquad | UCLA Library Data Science Center DataSquad

Paid students contribute to research and build RSE skills while they do it.



Campuses need more data and software help than staff can provide. Courses leave gaps. So does the code researchers write. DataSquads fill those gaps. Many students never hear of this work, especially students from underrepresented groups.

A paid job that teaches

Students work for the library or a department, and staff supervise them. Students help researchers with their data, from cleaning to reproducibility.

Who is on a DataSquad

Project manager	triages requests
Data analysts and developers	do the hands-on work with data and code
Technical writers	document projects
Staff	supervise, mentor, fund, and sustain the DataSquad

Limits we keep

We help with a researcher's data and code. We don't change their research questions, tutor their students, or share anything a researcher tells us in confidence.

What slows us down

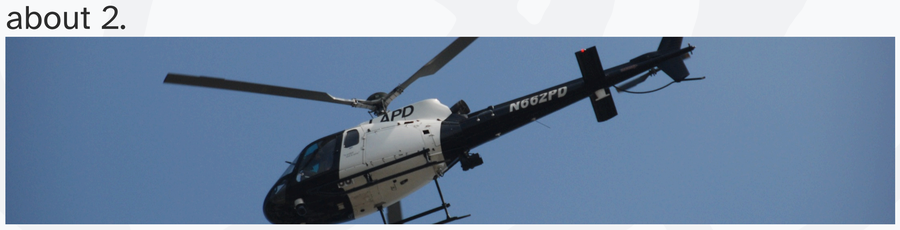
Student schedules, the academic calendar, onboarding, review, and graduation handoffs all slow the work. More students can take more varied projects, but they need more oversight. Handbooks, templates, and staff stay when students graduate.

What the work looks like

CARLETON Election-news scraper. Seven students built it in 2022. Two more extended it from 2023 to 2024 for a political science project that involved over 500,000 transcripts.

2022 ●●●●●●●
2023-24 ●●

UCLA Police helicopter flight data. Two students brought software skills to a research lab's R code. They refactored it with staff, and the code went through code review. Loading the data fell from about 318 minutes to about 2.



LAPD helicopter, for illustration. Photo: Chris Yarzab, CC BY 2.0, via Wikimedia Commons. Cropped and resized.

Before 318 min

After 2 min

Research software habits, when the project needs them. Run instructions, a dependency list, one test, and a citation file. We do not treat them as a checklist. We plan to track whether researchers keep them. wbcheck (**doi:10.5281/zenodo.23329531**) and the UC OSPO Education site (**doi:10.5281/zenodo.23330022**) have citation files that credit their student developers.

Where students go next

We located 19 former UCLA student employees. Their current job categories, from public profiles in 2026:

Data / analytics / consulting	**11**
Software / engineering / IT	**4**
Research / graduate school	**2**
Nonprofit / public sector	**2**

Descriptive, not a placement rate.

What adopting takes

- Student positions and a wage budget
- Supervisors and mentors with time to review and coach
- Onboarding and a home for documentation
- A plan for turnover

Start a DataSquad

Get the starter kit for roles, onboarding, confidentiality, and request tracking. Students learn software sharing from our open lesson, **Sharing Research Software**. Supervise students? Compare onboarding, review, and graduation handoffs with us.

datasquad.info/join